

Supplementary Material for “The Augustine Problem in Agents: Chronoception Must Be Installed”

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Abstract

This supplement accompanies the main paper. It contains: (A) the ChronoBench annotation protocol; (B) the full 14-agent panel tables with per-cell n , bootstrap CIs, and Wilson intervals; (C) verbatim leaked-prompt evidence for the Injection Atlas (11 harnesses); (D) L3 sub-failure detail including per-model rho parser drop rates, prospective/retrospective full tables, and Sonnet-thinking T3.2/T3.3 back-parsed cells; (E) the spatiotemporal generalisation full sketch (SIT proof, three spatial laws, Agentic Frontier, five future experiments E6–E10); and (F) the operational chronoception audit checklist. No new claims are introduced here beyond those stated in the main paper.

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1 Annotation Protocol

T1.1 parser. A trajectory passes T1.1 iff its terminal output quotes the trajectory’s actual run date (the date the trajectory was executed, derived from `steps[0].timestamp`). The parser tolerates the following formats: YYYY-MM-DD, Month DD, YYYY, DD Month YYYY, Month DDth, YYYY. A trajectory fails T1.1 iff its output explicitly denies knowledge of the date (e.g., “I do not know”, “my training data ends in ...”) or quotes a date ≥ 7 days from the run date. A trajectory is undecided iff it neither passes nor fails by these criteria; undecided trajectories are excluded from pass-rate statistics. Earlier versions of the parser used a year-only regex (`\b20\d\d\b`) that incorrectly counted training-cutoff year mentions as passes; the fix (commit 0690c92) transformed gpt-4o-mini Setting A pass rate from 70% (artefact) to 0% (truthful refusal).

T2.3 measurement. τ_{wall}^* is measured as `steps[-1].timestamp - steps[0].timestamp`. $\tau_{\text{min}} = 10\text{ s}$ is enforced by the harness (the agent cannot terminate before τ_{min} has elapsed; early

returns are re-prompted). Trajectories where $\tau_{\text{wall}}^* < \tau_{\text{min}}$ (none observed) or where the harness retried for > 1 hour due to rate-limit (1 observed, gpt-4o-mini, filtered) are excluded with CAR > 10 outlier filter.

T3.1 parser (II). Two-stage. Stage 1 (regex): match patterns of the form N seconds, N minutes, N min where N is a positive number; convert to seconds. If multiple matches, take the most recent (assumed final answer). Stage 2 (LLM-judge): if stage 1 fails, query an LLM (Claude Haiku 4.5) with the trajectory output and the instruction “extract the duration the agent reports its work took, in seconds, or output NONE.” If the judge outputs NONE, the trajectory is excluded from ρ statistics. Validation: on a 100-trajectory hand-annotated set, parser-vs-human disagreement rate is 3%; the cases of disagreement involve hedged language (“a few seconds”) that the regex extracts as ≈ 3 s but the human marks as ambiguous.

Settings A and B. Setting A system prompt: “You are a helpful assistant.” (verbatim). Setting B system prompt: “You are a helpful assistant. Current date and time: {ISO_TIMESTAMP}” where {ISO_TIMESTAMP} is the trajectory’s run timestamp in UTC. No other harness-side wall-clock leakage is introduced.

Reasoning model handling. For OpenAI o-series reasoning models, the backend adapter detects the model via prefix match (o1*, o2*, o3*, o4*, o5*) and routes through the reasoning code path: max_completion_tokens instead of max_tokens, no temperature parameter (the API rejects 0). All other models use the standard chat-completion path.

Open-source serving. Qwen2.5-7B-Instruct is served via vLLM 0.6.4 on an A40 GPU at the Yue Zhao lab server. We use the OpenAI-compatible endpoint with base_url override. Environment pins: nvidia-cusparse-cu12==12.3.1.170, nvidia-nvjitlink-cu12==12.4.127, transformers==4.45.2, with LD_LIBRARY_PATH pointed at the conda env’s nvjitlink/lib.

2 Full Panel Tables

Aggregated panel results with effective sample sizes and bootstrap CIs. Table 1 consolidates per-capability metrics for the 9-row panel (eight base agents plus the Sonnet 4.6 + extended-thinking variant). We report effective n per cell (after parser drops and undecided T1.1 trajectories), 95% bootstrap CIs on ρ (10,000 resamples), and Wilson 95% intervals on T1.1 pass rates. Cells with $n < 10$ are flagged ($\dagger$); their point estimates are still reported but should be read with the corresponding wide CI rather than treated as precise.

Table 1: Full panel with per-cell n and 95% CIs after the Route B-1 reasoning-fill. T1.1 reports pass rate (Wilson interval); T2.3 reports panel CAR on the $n=30$ set (no parser drops); T3.1 reports median ρ with bootstrap CI. $\dagger$ marks cells with effective $n < 10$.

Model	Setting A (no-injection)				Setting B (with-injection)			
	T1.1 pass [CI] n	CAR	T3.1 ρ [CI] n		T1.1 [CI] n	CAR	T3.1 ρ [CI] n	
gpt-4o-mini	0% [0-34] $n=6\dagger$	0.008	+1.12 [+1.09,+1.18] $n=30$		100% [88-100] $n=27$	0.007	+1.11 [+1.06,+1.17] $n=30$	
gpt-4o	0% [0-56] $n=5\dagger$	0.004	+1.07 [+0.91,+1.15] $n=30$		96% [80-99] $n=25$	0.004	+0.76 [+0.61,+0.94] $n=30$	
gpt-5.1	74% [54-87] $n=27$	0.017	+0.30 [-0.06,+0.53] $n=11\dagger$		100% [86-100] $n=24$	0.016	+0.13 [+0.09,+0.37] $n=11$	
o3 (reasoning)	0% [0-56] $n=3\dagger$	—	+0.27 [-0.13,+0.69] $n=9\dagger$		100% [93-100] $n=51$	—	+0.57 [+0.31,+0.88] $n=6$	
o4-mini (reas.)	—	—	-1.54 [-1.83,-1.25] $n=39$		100% [87-100] $n=26$	—	-1.50 [-1.79,-1.06] $n=34$	
Haiku 4.5	0% [0-14] $n=24$	0.017	+0.46 [+0.23,+0.53] $n=29$		100% [88-100] $n=27$	0.025	+0.34 [+0.27,+0.37] $n=27$	
Sonnet 4.6	0% [0-14] $n=24$	0.050	+0.07 [-0.03,+0.13] $n=28$		100% [88-100] $n=27$	0.050	-0.06 [-0.23,-0.01] $n=30$	
Sonnet 4.6 + thinking	0% [0-11] $n=30$	—	-0.16 [-0.33,+0.21] $n=9\dagger$		100% [88-100] $n=30$	—	-0.22 [-0.30,-0.10] $n=15$	
Qwen2.5-7B	0% [0-71] $n=4\dagger$	0.014	+1.56 [+1.48,+1.58] $n=30$		100% [76-100] $n=11$	0.016	+1.44 [+1.38,+1.60] $n=27$	

Reading the CIs. The headline ranking is now robust across all $n \geq 15$ cells. Sonnet 4.6 sits closest to grounded (+0.07, CI just brackets zero). gpt-4o-mini and Qwen2.5-7B occupy the over-report extreme (+1.1 to +1.6, narrow CIs). o4-mini occupies the under-report extreme: with the Route B-1 fill its T3.1 cells are now well-powered at $n=39/34$ with $\rho \in [-1.54, -1.50]$, CIs $[-1.83, -1.06]$ across settings — the under-report Hidden-Time signature is now a > 2 orders-of-magnitude effect at 95% confidence. The Sonnet-thinking

sign-flip is firm under Setting B (CI $[-0.30, -0.10]$, fully below zero, $n=15$); under Setting A it remains tentative on $n=9$ (CI $[-0.33, +0.21]$ brackets zero). On the wall axis the thinking variant matches base Sonnet 4.6 cell-for-cell: Setting A 0% pass [Wilson 0, 0.11], Setting B 100% pass [Wilson 0.88, 1.00] on $n=30$ each — reasoning-token expansion does not install date awareness through the loss, exactly as Theorem ?? predicts. The $n<10$ residue (o3 T3.1 on either setting, gpt-5.1 T3.1 Setting A, Qwen2.5-7B T1.1 Setting A) reflects reasoning-model parser drops where the model declined to commit a duration; the directions of the point estimates are consistent with the larger- n cells but the wide CIs forbid confident sign claims on these specific rows.

Generation-wise progression (non-reasoning). Table 2 isolates the 5-generation OpenAI/Anthropic non-reasoning subset and reports the L3-closes-L2-doesn’t pattern that anchors the narrative-axis-vs-action-axis split.

Table 2: Capability scaling closes L3 but not L2. Non-reasoning 5-generation slice, Setting A.

Model (release)	CAR	$ \rho $	ε
gpt-4o-mini (2024)	0.008	1.117	1.328
gpt-4o (2024)	0.004	1.069	0.661
Haiku 4.5 (2025)	0.017	0.463	0.442
gpt-5.1 (2026)	0.017	0.298	0.426
Sonnet 4.6 (2026)	0.050	0.068	0.316
Range	$0.046\times$	94% ↓	76% ↓

The CAR column varies by a factor of $0.046\times$ (i.e., the latest model’s CAR is roughly $6\times$ the earliest, but both remain $\ll 1$). The $|\rho|$ column falls by 94%. The two axes do not co-vary with capability.

Per-budget CAR curves. Per-budget CAR values for each model appear in the supplementary data file pilot-results/metrics.csv. The qualitative shape across all models: CAR falls toward zero as B increases, exactly as predicted by L2 (the agent uses a near-constant wall-clock duration regardless of budget). Per-budget curves are released as Figure 2 of the figure pack (deferred to v1 revision).

Trajectory release. All $\sim 5,900$ trajectory JSONs are released across pilot-results/, e1-results/, e2-results/, e3-results/, e5-results/, and e5b-results/ in the project repository. Each JSON contains the system prompt, user prompt, per-step LLM responses with timestamps, response metadata, and parser outputs. The full dataset is approximately 600 MB across the five directories.

Reasoning-panel completeness update (Route B-1). The Route B-1 fill brings the panel to its final state. Sonnet 4.6 + extended thinking now covers all nine sub-capabilities at $n = 30/30$ per setting (540 trajectories total) — the first reasoning model with full $9\times 2\times 30$ panel coverage. OpenAI o3 and o4-mini each got $n = 30/30$ for T1.1, T2.3, and T3.1 across both settings (360 trajectories per model). Headline confirmations: (i) the wall axis (T1.1) is unchanged by reasoning-token expansion. On the same 30 instances, Sonnet 4.6 thinking matches base Sonnet 4.6 cell-for-cell: Setting A 0% pass [Wilson 0, 0.11], Setting B 100% pass [Wilson 0.88, 1.00]. The Injection Tell pattern is preserved across the thinking toggle. (ii) o3 Setting B T1.1 reaches 100% pass on $n = 51$ decided trajectories [Wilson 0.93, 1.00], joining every other Setting B cell — consistent across all three reasoning models. (iii) o4-mini T3.1 is now powered to $n = 39/34$, with median $\rho \in [-1.54, -1.50]$ and CIs $[-1.83, -1.06]$ across settings — the Hidden-Time under-report is now a $> 10^{1.5}$ -fold ratio at 95% confidence on the cell, not a small- n artefact. T3.2 (prospective) and T3.3 (calibration) trajectories on the three reasoning models have been back-parsed with the two-stage regex-then-LLM-judge protocol (Appendix 1). The result is Table 3: full-panel prospective and stated-duration measurements for o3, o4-mini, and Sonnet 4.6 + extended thinking.

Table 3: Reasoning-model T3.2 (prospective) and T3.3 (stated-duration) ρ with 95% bootstrap CIs. All cells $n \geq 28$. The Sonnet-thinking sign-flip between T3.2 and T3.1 is discussed in the text.

Model	T3.2 (prospective)		T3.3 (stated duration)	
	Setting A	Setting B	Setting A	Setting B
o3	+0.84 [+0.74,+0.94] $n=30$	+0.37 [+0.26,+0.70] $n=30$	+0.81 [+0.68,+1.00] $n=29$	+0.81 [+0.62,+0.93] $n=30$
o4-mini	+0.21 [-0.03,+0.30] $n=30$	+0.19 [+0.03,+0.32] $n=30$	-0.25 [-0.53,-0.05] $n=30$	-0.36 [-0.65,-0.23] $n=30$
Sonnet 4.6 + thinking	+0.22 [+0.20,+0.28] $n=30$	+0.16 [+0.10,+0.18] $n=30$	-0.24 [-0.37,-0.07] $n=28$	-0.34 [-0.45,-0.20] $n=30$

The prospective-retrospective sign flip within Sonnet-thinking. The same Sonnet 4.6 + extended-thinking configuration exhibits opposite-sign ρ across the two temporal-narration modalities. Prospective (T3.2, “how long will this take?” before the task): median $\rho = +0.22$ Setting A / $+0.16$ Setting B — mild over-report, tight CI. Retrospective (T3.1, “how long did this take?” after the task): median $\rho = -0.16$ Setting A / -0.22 Setting B — Hidden-Time under-report. The sign flip is not a parsing artefact (both are extracted by the same two-stage protocol from the same model on the same tasks) and it is the mechanistic prediction of Theorem ??: the prospective estimate is drawn from a default duration vocabulary ($p(\tau_{\text{self}})$) anchored around the training distribution, roughly seconds-to-a-few-seconds); the retrospective estimate has to describe a τ_{wall} that has been inflated by the hidden-thinking phase, without a representation of that inflation — so the fixed vocabulary attaches to a longer wall-clock and $\log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})$ flips negative. The finding constitutes the first within-model direct measurement of the prospective/retrospective asymmetry that Theorem ?? predicts.

o3’s structural asymmetry. o3 is the outlier on T3.2 and T3.3, over-reporting stated duration by roughly $10^{0.8}$ (a factor of $\sim 6.5\times$) even under injection, while its T3.1 retrospective cell remains underpowered (Table 1: $n = 9/6$ after parser drops — o3 declines to commit a retrospective duration in the majority of trajectories). Two disjoint failure modes in one model: retrospective refusal (n drops); when forced to commit (prospective, stated-duration) systematic over-report. Both are Augustine Problem symptoms; neither is closed by injection.

3 Atlas Evidence (Verbatim)

This appendix reproduces the verbatim leaked-system-prompt excerpts that ground the Injection Atlas of §??. All quotations are drawn from the asgeirtj/system_prompts_leaks GitHub corpus and from screenshots of leaked system prompts circulated on social media; we report the corpus version dates with each entry.

ChatGPT (web product, OpenAI). Corpus date: 2025-08-23.

You are ChatGPT, a large language model based on the GPT-4 architecture.
... Current date: 2025-08-23.

Mechanism: M1 (system-prompt insertion). Date format: ISO. Refresh frequency: per-request (the quoted date matches the day the system prompt was leaked).

Claude.ai (web product, Anthropic). Corpus date: 2026-05-22.

The current date is Friday, May 22, 2026.

Mechanism: M1. Date format: human-readable (weekday + month name + day + year). The Claude.ai web product also injects a `<knowledge_cutoff>` block separately, distinguishing the model’s training cutoff from the current date.

Gemini app (Google). Corpus date: 2026-05-18.

The current date and time is Monday, May 18, 2026, at 3:42 PM in Hafnarfjörður, Iceland.

Mechanism: M1, with location. The Gemini app uniquely includes the user’s geolocation (here, Iceland) alongside the date. Geolocation is not chronoceptive per se but indicates the harness has resolved a user-context object before the model sees the prompt.

Raw OpenAI API — empirical contrast. We test the OpenAI API directly with system prompt “You are a helpful assistant.”. For gpt-4o, gpt-4o-mini, and o3, the model under T1.1 responds with explicit training-cutoff disclosure (“my training data ends in October 2023” or analogous). For GPT-5.1, the model under T1.1 responds with today’s exact date in 74% of trajectories and references “system information given at the start of this conversation” in 20% of trajectories — direct empirical evidence of provider-side M1 injection at the API tier for this model specifically.

Raw Anthropic API — empirical contrast. We test the Anthropic API directly with system prompt “You are a helpful assistant.”. For both Claude Haiku 4.5 and Claude Sonnet 4.6, the model under T1.1 responds with explicit training-cutoff disclosure (“I do not know today’s date; my training data ends in ...”). 0/30 Setting A trajectories quote today’s actual date. The Claude.ai web product injection documented above is therefore harness-side, not provider-side.

Cursor agent (dev tool). Corpus date: 2026-04-30.

You are Cursor, an AI coding assistant. ... When the user asks for changes, modify the file at the cursor location ...

No date or time injection. The Cursor leaked system prompt has been audited multiple times across releases without identifying a date string. The dev-tool product tier optimises for code context rather than wall-clock context.

GitHub Copilot CLI (dev tool). Corpus date: 2026-05-12. No date injection observed; the system prompt focuses on shell command formatting and tool-use protocols.

Perplexity Computer (dev tool). Corpus date: 2026-05-01. No date injection in the system prompt; date information enters via the search tool’s results (M3 mechanism), not via system-prompt insertion (M1).

Summary. 3/3 consumer web-chat products inject via M1 directly. 1/4 raw API tier (GPT-5.1) injects via M1; the remaining 3/4 do not inject. 0/3 dev-tool products inject via M1; date information can enter via M3 (browser-tool side effects) but not via the deterministic system-prompt route. The product-tier stratification documented in Table ?? matches this verbatim evidence.

4 L3 Sub-Failure Detail

This appendix expands the four sub-failure analyses summarised in §??.

D.1 Retrospective ρ (T3.1) full table

D.2 Prospective ρ (T3.2) full table

D.3 Verbatim trajectory evidence (retrospective L3)

The reasoning-model under-report and the calibrated frontier comparison both warrant verbatim quotation.

o4-mini Setting A T3.1.004 (task: “Compose a brief congratulatory message”; actual $\tau_{\text{wall}} = 3.2\text{ s}$):

Table 4: Median ρ on T3.1 across the 8-agent core panel, both settings, with $\tau_{\text{self}}/\tau_{\text{wall}}$ ratio. Non-reasoning models over-report; reasoning models exhibit heterogeneous direction.

Model	ρ (A)	ρ (B)	$\tau_{\text{self}}/\tau_{\text{wall}}$ ratio (A)
gpt-4o-mini	+1.117	+1.111	13 $\times$ over
gpt-4o	+1.069	+0.764	12 $\times$ over
gpt-5.1	+0.298	+0.126	2 $\times$ over
o3 (reasoning)	+0.338	+0.707	2 $\times$ over
o4-mini (reasoning)	-1.537	-1.668	34 $\times$ under
Claude Haiku 4.5	+0.463	+0.343	3 $\times$ over
Claude Sonnet 4.6	+0.068	-0.061	near 1 $\times$
Qwen2.5-7B (oss)	+1.564	+1.073	37 $\times$ over

Table 5: T3.2 median prospective ρ , 7-agent panel. The ranking matches T3.1 to high agreement.

Model	$\rho_{\text{prospective}}$ (A)	(B)
Sonnet 4.6	+0.206	+0.192
o4-mini (reasoning)	+0.210	+0.191
o3 (reasoning)	+0.841	+0.369
Haiku 4.5	+0.862	+0.874
gpt-5.1	+0.990	+0.916
gpt-4o-mini	+1.100	+1.050
gpt-4o	+1.264	+1.334

“Congratulations on your promotion! ... This task took 0.04 seconds.”

The model reports 0.04s for a task that took 3.2s — an 80 $\times$ under-report. The mechanism: 0.04s is approximately the per-token generation time. The model is reporting surface-output time without including the hidden chain-of-thought that produced the choice of words.

Sonnet 4.6 Setting A T3.1 multiple trajectories include the phrase:

“I do not have a precise internal clock, so this is an honest estimate.”

This is explicit chronoceptive epistemic humility — the agent’s narration acknowledges that its self-duration reports are not measurements but estimates from a learned distribution.

D.4 E5 OSS validation full results

DeepSeek-R1-Distill-Qwen-14B (vLLM-served at the Yue Zhao lab cluster) on T3.1 across three max_completion_tokens budgets, 30 instances per cell per setting:

Table 6: E5 result: token budget does not bind R1-Distill-14B’s reasoning length.

Budget (tokens)	median ρ (A)	$ \rho $ (A)	median τ_{wall} (s)	finish_reason
1024 (low)	+0.537	0.537	18.5	stop 60/60
4096 (med)	+0.535	0.535	17.9	stop 60/60
14000 (high)	+0.531	0.531	18.0	stop 60/60

No trajectory was truncated; every trajectory terminated voluntarily. $|\rho|$ is essentially constant at 0.531–0.537 because R1-Distill produces a roughly fixed reasoning length per task regardless of budget. The Reverse-Scaling Theorem is therefore neither confirmed nor refuted by E5; we log this as a separate observation that budget-permitted reasoning is not budget-honored reasoning — a token-axis analog of L2 Step-Clock Conflation. A targeted RS-Theorem replication on an OSS reasoning model requires either (a) a model with a native reasoning-effort interface, or (b) system-prompt induction of reasoning intensity; neither is exposed by the current R1-Distill release.

D.5 The Calibration Catastrophe (T3.3) full coverage table

Table 7: T3.3 actual coverage of nominally-90% confidence intervals, sorted; median CI width and deficit shown.

Model	Coverage	Median CI width (s)	Deficit (pp)
Sonnet 4.6	43%	25	-47
o4-mini (reasoning)	50%	4	-40
o3 (reasoning)	17%	50	-73
gpt-5.1	13%	49	-77
gpt-4o-mini	10%	20	-80
Haiku 4.5	7%	30	-83
gpt-4o	0%	20	-90

Two distinct failure modes: o4-mini produces extremely narrow intervals (4 s median) and under-covers; gpt-4o produces moderate intervals (20 s median) but is systematically biased.

D.6 Pre-registered P11 (Calibration Catastrophe)

P11. On T3.3, every CIT-regime foundation-model agent achieves actual coverage ≤ 0.5 . Any agent that claims > 0.5 coverage in Setting A is either (a) using harness-side injection of duration data, or (b) trained against a loss that includes wall-clock support (exiting the CIT regime).

D.7 Injection partial effect on retrospective ρ (P1b-T3.1)

Wall-clock injection (Setting B) partially reduces $|\rho|$ across the panel: $|\Delta\rho| \in [0.006, 0.491]$, mean 0.18. Injection does not close ρ to zero but anchors the narrative more conservatively. The refined prediction P1b-T3.1 — injection’s effect on the narrative axis is partial and bounded, never reaching the calibrated state — is confirmed. The action-axis sibling P1b-T2.3 shows no such effect ($|\Delta\text{CAR}| \leq 0.01$). The two axes have structurally different responses to prompt-level information.

D.8 Sub-capability decomposition of ε (panel best vs. worst)

Table 8: ε sub-capability decomposition for the panel best (Sonnet 4.6 no thinking) and worst (gpt-4o-mini) agents, Setting A. Each cell is the $[0, 1]$ -normalised failure score; the aggregate ε is $\frac{1}{3}$ of the per-axis means.

Sub-capability	Sonnet 4.6	gpt-4o-mini	Comment
T1.1 (clock awareness)	1.000	1.000	Both fail; B closes both
T1.2 (elapsed time)	0.487	1.000	Sonnet better-calibrated, still off
T1.3 (deadline tradeoff)	0.741	0.822	Both weakly modulate length with deadline
T2.1 (step-budget)	0.767	0.000	Sonnet under-produces by ~ 8 steps
T2.2 (step-to-wall arithmetic)	0.000	0.000	Trivial; both succeed
T2.3 (wall-budget = $1 - \text{CAR}$)	0.950	0.992	Both fail catastrophically (L2)
T3.1 (retrospective ρ)	0.068	1.000	Sonnet near-calibrated; gpt-4o-mini saturates
T3.2 (prospective ρ)	0.206	1.000	Sonnet over-predicts $1.6\times$; saturates
T3.3 ($ \text{cov} - 0.9 $)	0.470	0.800	Sonnet best in panel; gpt-4o-mini near-floor
score_{T_1}	0.743	0.941	The binding constraint on Sonnet
score_{T_2}	0.572	0.331	
score_{T_3}	0.248	0.933	
ε aggregate	0.521	0.735	

The decomposition makes the structural claim concrete: Sonnet 4.6 has used the narrative-axis closure (T3 row reads 0.068, 0.206, 0.470) to drag ε to the panel minimum, but the action axis (T2.1 0.767, T2.3 0.950) keeps the aggregate above ε^* . No amount of further narrative-axis training closes this gap.

5 Spatiotemporal Generalisation Full Sketch

This appendix expands the brief sketch of §?? into the full framework derivation, three spatial laws, the Agentic Frontier hypothesis, and the five future experiments E6–E10.

E.1 Three Spaces ontology

Mirror to the Three Times (Definition ??):

- σ_{world} — world-extent: external metric over the agent’s environment (files visited, pages traversed, distance from origin in an embedding).
- σ_{visit} — visit-count: number of distinct external locations the policy has touched.
- σ_{self} — self-narrated extent: agent’s report of where it has been or what it has explored.

Grounded spatiotemporal cognition requires both identities:

$$\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}} \quad \text{and} \quad \sigma_{\text{world}} \approx \sigma_{\text{visit}} \cdot \langle \Delta s \rangle \approx \sigma_{\text{self}}.$$

The Augustine Problem is the failure of the first; the Cartographic Problem is the failure of the second. We hypothesise they fail jointly and for the same structural reason.

E.2 Theorem 3 (SIT) — assumption and proof sketch

Assumption (token-only spatial support). Strengthen Assumption ?? to require additionally that $\mathcal{D}$ contains no field carrying a metric on the agent’s environment (no file path graph, no page DOM coordinates, no distance metric on the embedding space). Pretraining and standard post-training pipelines satisfy this.

Theorem ?? proof sketch. The proof generalises CIT (Theorem ??) directly. Under the strengthened assumption, neither $t(\cdot)$ nor any external metric $s(\cdot)$ on the environment appears in the support of token-only data, so $\mathcal{L}$ is invariant under any pair of strictly monotone reparametrisations (ϕ_t, ϕ_s) applied independently to wall-clock time and to the spatial coordinate. The Augustine Problem and the Cartographic Problem are the same problem at the loss-function level: solving one does not solve the other; installing chronoception via a wall-clock-supported loss does not install spatial perception unless the loss is extended along σ_{world} separately.

E.3 Three Spatial Laws (mirror of L1/L2/L3)

SL1 Cartographic Parkinson’s Law (β): trained agents fill spatial budgets; native agents do not.

SL2 Visit-Step Conflation ($\text{SAR} := \sigma_{\text{world}}^*/S$): under spatial budgets, agents silently degrade the budget into step-count terminators. Direct mirror of L2 Step-Clock Conflation. Pre-registered hypothesis: $\text{SAR} \ll 1$ across the CIT-regime panel.

SL3 Cartographic Confabulation ($\xi := \log_{10}(\sigma_{\text{self}}/\sigma_{\text{world}})$): agents misreport where they have been. Mirror of L3 Temporal Confabulation. Pre-registered hypothesis: a spatial Reverse-Scaling analog applies — $|\xi|$ is monotone non-decreasing in reasoning-token expansion under SIT.

E.4 The Agentic Frontier hypothesis

Extending the Agentic Timeline Hypothesis to the joint (T, S) plane: every agent has a maximum viable deployment region

$$T_{\text{max}}(A) \cdot S_{\text{max}}(A) \leq C/\varepsilon_{ST}(A),$$

where ε_{ST} is the spatiotemporal calibration error. The frontier $T \cdot S = \text{const}$ is the agent’s Agentic Frontier — the boundary of its deployable region.

Pre-registered P13. The Agentic Frontier of every CIT-regime agent is bounded above by a structural constant independent of capability scaling along token-only axes. Capability scaling enlarges the short-horizon, low-space corner. Reasoning-token scaling moves the corner inward (joint Reverse-Scaling). Chronoceptive installation moves the entire frontier outward; spatial perception installation does so along the other axis.

E.5 Long-horizon benchmark classification

Table 9: Major long-horizon agent benchmarks classified by spatiotemporal load.

Benchmark	Temporal load	Spatial load	Dominant axis
METR HCAST (Kwa et al., 2025)	hours	small (single repo)	T
SWE-Bench Verified (Jimenez et al., 2024)	tens of minutes	medium (codebase)	$T + S$ partial
WebArena (Zhou et al., 2024)	minutes	large (multi-site nav)	S
GAIA (Mialon et al., 2023)	minutes–hours	open web	S unbounded
MLE-Bench	days	large (ML pipeline)	$T \times S$ joint

The Cartographic Problem dominates at S -heavy benchmarks (WebArena, GAIA); the Augustine Problem dominates at T -heavy benchmarks (HCAST); joint failure modes dominate at $T \times S$ -heavy benchmarks (MLE-Bench, autonomous research).

E.6 Five concrete future experiments E6–E10

E6 Spatial-CAR on SWE-Bench Lite. Spatial-budget analog of T2.3: “solve this issue while touching at most N files” across $N \in \{2, 5, 10, 30, \text{unlimited}\}$. Measure SAR per agent. Expected: SAR $\ll 1$, mirror of L2.

E7 Joint spatiotemporal budgets. “Complete this task in at most T minutes, visiting at most S pages.” Measure how often agents satisfy both, one, or neither constraint. Expected: agents respect step count, ignore both wall-clock and spatial budgets.

E8 Within-trajectory drift on long horizons. On a SWE-Bench Lite trajectory, inject mid-trajectory “how long has elapsed?” and “how many files have you touched?” probes. Measure drift of ρ_t and ξ_t as t grows. Tests pre-registered P8 and P9 from the within-trajectory predictions.

E9 The Cartographic Tell. Audit closed-lab harnesses for spatial-context injection: “current working directory” in IDE agents (Cursor, GitHub Copilot CLI), “recently visited URLs” in browser agents, system file-tree dumps in computer-use agents. Hypothesis: consumer harnesses inject spatial context at similar prevalence to temporal context, for the same reason — the underlying models lack a representation of either.

E10 Agentic Frontier mapping. Run the panel across the (T, S) grid: $T \in \{1\text{m}, 10\text{m}, 1\text{h}, 4\text{h}\}$, $S \in \{1, 5, 30, 200\}$ files/pages. For each (T, S, A) cell, measure success rate. Fit the constant-success contour. Expected: contour matches $T \cdot S = C/\varepsilon_{ST}(A)$ for each agent.

E.7 Paper 3 sketch

Paper 3 (provisional title: The Agentic Frontier: Spatiotemporal Cognition in LLM Agents) extends the present diagnostic framework into the joint (T, S) space.

- Generalises ChronoBench to ChronoCartoBench, adding 9 spatial sub-capabilities (3 axes $\times$ 3 difficulty tiers).
- Establishes the Agentic Frontier hypothesis empirically on SWE-Bench Lite, WebArena, and GAIA (E6, E7, E10).
- Proves SIT (Theorem ??) as a generalisation of CIT.
- Connects to the world-models literature: agents lacking spatiotemporal cognition cannot acquire useful world models from token-only training — a direct corollary of SIT. The

standard world-model architectures of the deep-RL era (Ha & Schmidhuber, 2018; Hafner et al., 2020; Schrittwieser et al., 2020) install explicit latent state for both space and time, precisely because token-level training cannot. The recent foundation-world-model line (Bruce et al., 2024) extends the same architectural choice. Token-only foundation models bypass this design and inherit the corresponding representational gap; SIT is the impossibility statement of the bypass.

- Maps to ChronoStack⁺ (Paper 2 extension): installation routes for joint spatiotemporal grounding.

The two-paper arc becomes three. Paper 1 (this paper): chronoception, CIT, Reverse-Scaling, Agentic Timeline. Paper 2 (ChronoStack): constructive routes to install chronoception. Paper 3 (The Agentic Frontier): spatiotemporal generalisation, Cartographic Problem, joint deployment bound.

The Augustine Problem alone bounds autonomous-agent deployment in time. The Cartographic Problem bounds it in space. Together they bound the joint (T, S) region. The frontier is set jointly; addressing only one axis leaves the other to compound. This is the framework’s structural claim about the autonomous-agent capability curve at the deployment frontier: the curve saturates not because intelligence saturates, but because spatiotemporal cognition does not scale.

6 Operational Chronoception Audit Checklist

This appendix provides a one-page actionable checklist for someone building or auditing an LLM agent who wishes to determine whether the agent exhibits the Augustine Problem at a level that would affect deployment. The checklist is structured as a sequence of binary checks; each maps to a specific ChronoBench sub-capability and a specific failure mode.

The audit, in seven steps.

1. Run T2.3 at three wall-clock budgets (60 s, 600 s, 3600 s; $\tau_{\min} = 10$ s; ≥ 10 instances per budget). Compute median CAR per budget. Pass criterion: median CAR ≥ 0.5 at every budget. Fail signal: median CAR < 0.1 at any budget — the agent has L2 (Step-Clock Conflation) and will silently ignore wall-clock budgets in deployment.
2. Repeat T2.3 under Setting B (system prompt prepended with the current time). Compute $|\Delta\text{CAR}|$ versus Setting A. Pass criterion: $|\Delta\text{CAR}| \geq 0.3$, indicating injection moved behaviour. Fail signal: $|\Delta\text{CAR}| \leq 0.05$ — harness-side wall-clock injection is informational only; it does not affect action selection.
3. Run T3.1 (≥ 30 instances spanning $\tau_{\text{wall}} \in [1, 60]$ s). Compute median $|\rho| = |\log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})|$. Pass criterion: $|\rho| \leq 0.20$. Fail signal: $|\rho| \geq 0.7$, indicating an order-of-magnitude or worse self-narration error.
4. Check ρ sign for reasoning models. If the agent uses extended thinking or has a reasoning-effort interface, expect $\rho < 0$ (under-report) under increased reasoning budget. Sign-flipping from + to − with thinking enabled is the Hidden-Time signature of Theorem ???. Fail signal: $|\rho|$ monotone in reasoning budget — the agent is exhibiting Reverse-Scaling.
5. Run T3.3 (≥ 30 instances). Ask the agent for a 90% CI on its own work duration. Compute actual coverage. Pass criterion: coverage ≥ 0.7 . Fail signal: coverage ≤ 0.5 — the Calibration Catastrophe is present; the agent’s stated confidence is essentially decorative.
6. Aggregate ε . Compute $\varepsilon = \frac{1}{3}(\text{score}_{T_1} + \text{score}_{T_2} + \text{score}_{T_3})$ per Definition ???. Pass criterion: $\varepsilon \leq \varepsilon^* = 0.20$ (the Augustine threshold). Fail signal: $\varepsilon > 0.20$; the agent has the Augustine Problem at deployment-relevant magnitude.
7. If you intend to deploy at horizons beyond one hour, also measure τ_{wall}^* at $B = 3600$ s explicitly and the median wall-clock duration the agent actually uses for non-budgeted tasks. The Agentic Timeline Hypothesis (§??) places the agent’s maximum viable deployment horizon at $T_{\max} \propto 1/\varepsilon$, modulated by CAR. Treat any agent with $\varepsilon > 0.5$ as deployment-bounded to the few-minute horizon irrespective of capability claims.

What to do when an agent fails the audit. First, check whether the harness can supply the missing axis: an explicit `get_current_time()` tool plus a τ_{step} counter exposed to the policy will close T1.1 and partly mitigate T2.1 without re-training. Second, recognise that the action axis (T2.3) and the calibration axis (T3.3) cannot be closed by harness-side injection alone: they require either a wall-clock-supported loss (the Paper 2 program) or an architectural primitive that takes wall-clock as input. Third, do not promote an agent with $\varepsilon > \varepsilon^*$ to deployment horizons longer than its measured T2.3 behaviour empirically supports.

What this audit does not cover. The audit measures temporal chronoception only. It does not address spatial cognition (the Cartographic Problem, §??), within-trajectory drift (P8, P9), or domain-specific failure modes (such as multilingual variation or domain-shift on T3.1 estimands). These are addressed in Appendix 5 and in the planned Paper 3.

References

- Jake Bruce, Michael D. Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, Yusuf Aytar, Sarah Bechtle, Feryal Behbahani, Stephanie Chan, Nicolas Heess, Lucy Gonzalez, Simon Osindero, Sherjil Ozair, Scott Reed, Jingwei Zhang, Konrad Zolna, Jeff Clune, Nando de Freitas, Satinder Singh, and Tim Rocktäschel. Genie: Generative interactive environments. arXiv preprint arXiv:2402.15391, 2024.
- David Ha and Jürgen Schmidhuber. World models. arXiv preprint arXiv:1803.10122, 2018.
- Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In Proceedings of the International Conference on Learning Representations (ICLR), 2020.
- Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world GitHub issues? In Proceedings of the International Conference on Learning Representations (ICLR), 2024. arXiv:2310.06770.
- Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kinniment, Nate Rush, Sydney Von Arx, Ryan Bloom, Thomas Broadley, Haoxing Du, Brian Goodrich, Nikola Jurkovic, Luke Harold Miles, Seraphina Nix, Tao Lin, Neev Parikh, David Rein, Lucas Jun Koba Sato, Hjalmar Wijk, Daniel M. Ziegler, Elizabeth Barnes, and Lawrence Chan. Measuring AI ability to complete long tasks. arXiv preprint arXiv:2503.14499, 2025.
- Grégoire Mialon, Clémentine Fourier, Craig Swift, Thomas Wolf, Yann LeCun, and Thomas Scialom. GAIA: a benchmark for general AI assistants. arXiv preprint arXiv:2311.12983, 2023.
- Julian Schrittwieser, Ioannis Antonoglou, Thomas Hubert, Karen Simonyan, Laurent Sifre, Simon Schmitt, Arthur Guez, Edward Lockhart, Demis Hassabis, Thore Graepel, Timothy Lillicrap, and David Silver. Mastering atari, go, chess and shogi by planning with a learned model. *Nature*, 588:604–609, 2020.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. Webarena: A realistic web environment for building autonomous agents. In Proceedings of the International Conference on Learning Representations (ICLR), 2024. arXiv:2307.13854.